

Ahmed Mostafa

2141 Virginia St, Berkeley, CA, United States

(949) 426-5899 | ahmostafa@berkeley.edu | linkedin.com/in/ahmed-mostafa147 | github.com/ahmostafa147

EDUCATION

University of California, Berkeley

Aug. 2025 - May 2026

M.Eng. in Electrical Engineering & Computer Science

University of California, San Diego

Sep. 2022 - March 2025

B.S. in Data Science, Minor in Mathematics

GPA: 4.0/4.0

Relevant Coursework:

Data Structures & Algorithms, Machine Learning, Systems for Scalable Analytics, Agentic AI, Responsible Data Science

EXPERIENCE

Machine Learning Intern

June 2024 – Sep. 2024

CarsXE

Highland, MD

- Deployed regression car valuation model aiding customer assessment of vehicle prices, using Streamlit and Sklearn
- Developed automated pipelines integrating multiple **RESTful APIs**, streamlining data for model training
- Performed exploratory data analysis, reducing feature space and memory usage by **54%**, using Seaborn and Pandas

Software Engineering Intern

June 2023 – Sep. 2023

CarsXE

Highland, MD

- Developed **full-stack** user monitoring system using Python, implementing analytics to identify growth opportunities
- Engineered API data validation step, enhancing reliability through systematic consistency checks and error handling
- Optimized backend performance by caching dynamic queries, reducing dashboard runtime from **15 min to seconds**

Instructional Assistant

Sep. 2023 – March 2025

UC San Diego – Halicioglu Data Science Institute

San Diego, CA

- Tutored **1000+** students, improving understanding of different machine learning models through discussion sessions
- Utilized students tutoring session feedback, improving **Java**, Data Structures, and **debugging** knowledge retainment
- Created **version control system** project, aiding students to understand Git using simplified write-up approaches

PROJECTS

Radiographic Enhanced Sepsis Detection System — *Python, TensorFlow, AWS, CatBoost, React*

- Utilized chest Xrays & lab test to train an early sepsis detection model achieving **0.88 AUC** using SageMaker
- Processed and integrated MIMIC-IV and MIMIC-CXR datasets using SQL and probabilistic imputation techniques
- Engineered ResNet-50 CNN model achieving **96% recall** for pneumonia by fine-tuning transfer learning architecture
- Deployed model using AWS ECS, automating updates via EventBridge and Lambda for smooth integration

Sepsis Clustering in ICU Patients | *R, PostgreSQL, Docker, Git*

- Replicated a research paper on multimorbidity & sepsis in ICU patients, analyzing **36,000+** EHRs of MIMIC-III
- Trained latent class analysis to identify patient subgroups, connecting different risks for organ dysfunction and sepsis
- Created network visualization to highlight critical multimorbidity patterns, aiding in subgroups labeling, using ggraph
- Documented reproducibility processes in GitHub, ensuring robust methodologies tracking in the MIMIC-III analysis

League of Legends: Winning Prediction | *Sci-kit learn, Pandas, Numpy, Plotly*

- Implemented a **Random Forest Classifier** to predict match outcomes, using **GridSearchCV** for parameter tuning
- **Achieved a 34% increase to prediction accuracy** by integrating champion-specific statistics and team dynamics
- Discovered that in lane to lane comparisons, support roles have an inverse relationship between kills and win-rate
- Conducted **permutation test** to evaluate model bias across team sides, assessing the fairness of the model

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript

Frameworks: Spark, Dask, TensorFlow, Streamlit, JUnit, GitHub Actions CI/CD

Developer Tools: Git, Docker, VS Code, IntelliJ, Firebase, Jira, Tableau

Libraries: Pandas, NumPy, Plotly, Sklearn, Seaborn, D3.js, Scipy, Keras